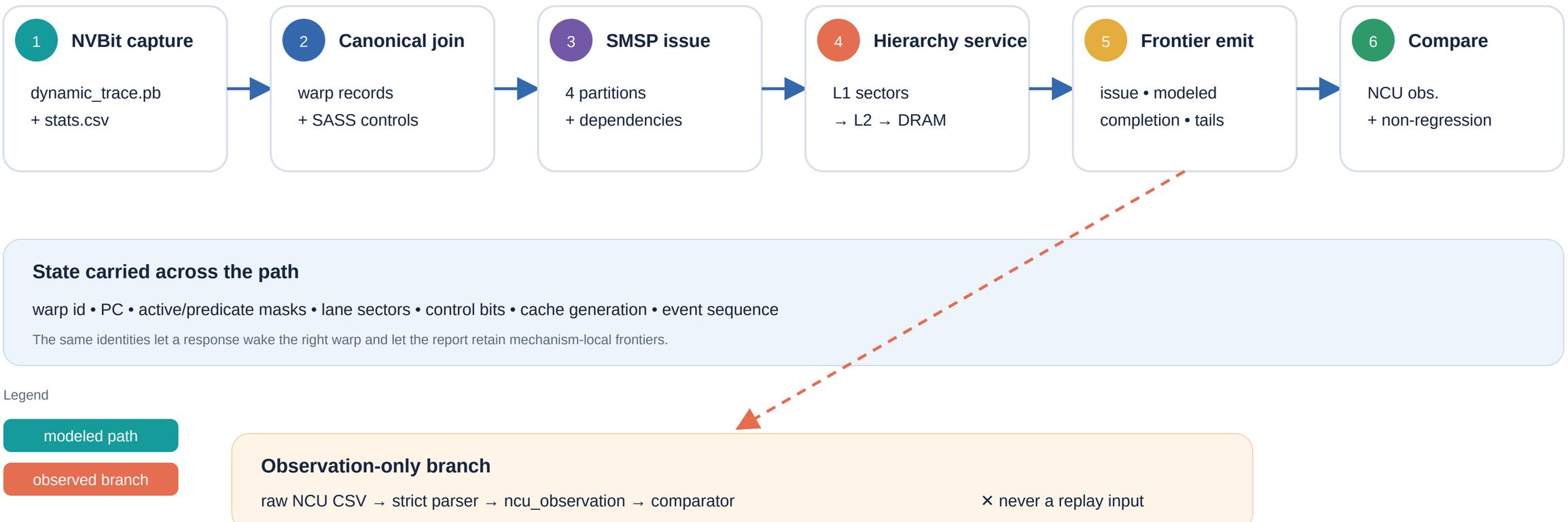
Six transformations turn a GPU trace into evidence

Each arrow names an artifact or invariant; the dotted branch never feeds replay



Caption. Numbered stages make the simulator’s contribution inspectable: concrete trace artifacts are normalized, scheduled, serviced, and emitted as separate frontiers; NCU is joined only after replay.

Source: sim/trace, sim/sm, sim/mem, sim/direct_run/src/ncu.rs; uPIMulator HPCA’24 and Centauri ASPLOS’24 figure grammar.